

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		121767	
Total Threats Mitigated	50	Total Packets Analyzed		6684304879	
High Priority	17	Malicious Packets		6657913111	
Low Priority	33	True Negatives Passed		94420	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		6	
ML Metrics	Value	SYN Flood		18	
False Positives	249	UDP Flood		12	
FP Rate	0.21%	Uncertain		14	
IF Precision	99.26%				
IF Recall	94.38%	Action		Count	
IF F1-Score	96.76%	Quarantined		11	
IF Accuracy	98.57%	Time Ban		10	
		Blackhole		0	
		Released		8	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.26%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	94.38%	Share of actual attacks successfully flagged
F1-Score	96.76%	Balanced measure combining Precision and Recall
Accuracy	98.57%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.21%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	5.62%	Actual attacks that went undetected
True Positive Rate (TPR)	94.38%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.79%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 25603	FN 1525
Actual: Normal	FP 191	TN 92895

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.97%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.99%	Balanced measure combining Precision and Recall
Accuracy	99.97%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	3.66%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	96.34%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	12870	0	0
Act: ICMP	0	3774	0
Act: UDP	0	0	8959

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.26%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	94.38%	Share of actual attacks successfully flagged
F1-Score	96.76%	Balanced measure combining Precision and Recall
Accuracy	98.57%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.21%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	5.62%	Actual attacks that went undetected
True Positive Rate (TPR)	94.38%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.79%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 25603	FN 1525
Actual: Normal	FP 191	TN 92895

2d. Response Latency

Metric	Value	Description
Detection Time	421.2 ms	Average time taken to detect an attack
Mitigation Response Time	9.7 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	2.54%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	13.10%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	5.92%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.16	3	2	2026-09-25 12:29:22	2026-09-25 12:30:43
10.0.0.17	2	2	2026-09-25 12:29:24	2026-09-25 12:30:35
10.0.0.18	2	2	2026-09-25 12:29:22	2026-09-25 12:30:33
10.0.0.19	2	2	2026-09-25 12:29:24	2026-09-25 12:30:35
10.0.0.20	1	1	2026-09-25 12:29:53	2026-09-25 12:30:03
10.0.0.22	1	1	2026-09-25 12:29:53	2026-09-25 12:30:04
10.0.0.23	1	1	2026-09-25 12:30:23	2026-09-25 12:30:33
10.0.0.24	1	1	2026-09-25 12:30:26	2026-09-25 12:30:36

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.25	1	1	2026-09-25 12:30:23	2026-09-25 12:30:33

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 12:06:02	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:02	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:03	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:03	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:12	10.0.0.18	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-25 12:06:12	10.0.0.16	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-25 12:06:12	10.0.0.16	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-25 12:06:12	10.0.0.18	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-25 12:06:13	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-25 12:06:13	10.0.0.19	DDoS	SYN Flood	92.5%	Critical	Time Ban (1m)
2026-09-25 12:06:13	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-25 12:06:13	10.0.0.19	DDoS	SYN Flood	92.5%	Critical	Time Ban
2026-09-25 12:06:33	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:33	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:34	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:43	10.0.0.20	DDoS	UDP Flood	75.6%	High	Time Ban (1m)
2026-09-25 12:06:43	10.0.0.20	DDoS	UDP Flood	75.6%	Critical	Time Ban
2026-09-25 12:06:44	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban (1m)
2026-09-25 12:06:44	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban (1m)
2026-09-25 12:06:44	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban
2026-09-25 12:06:44	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban
2026-09-25 12:07:03	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:07:03	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:07:05	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:07:12	10.0.0.18	DDoS	SYN Flood	92.7%	High	Released
2026-09-25 12:07:12	10.0.0.16	DDoS	SYN Flood	92.7%	High	Released
2026-09-25 12:07:13	10.0.0.17	DDoS	SYN Flood	93.3%	High	Time Ban (2m)
2026-09-25 12:07:13	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Released
2026-09-25 12:07:13	10.0.0.23	DDoS	ICMP Flood	90.8%	High	Time Ban (1m)
2026-09-25 12:07:14	10.0.0.23	DDoS	ICMP Flood	90.8%	Critical	Time Ban
2026-09-25 12:07:14	10.0.0.25	DDoS	ICMP Flood	91.2%	High	Time Ban (1m)
2026-09-25 12:07:15	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban
2026-09-25 12:07:15	10.0.0.24	DDoS	ICMP Flood	91.2%	Critical	Time Ban (1m)
2026-09-25 12:07:16	10.0.0.24	DDoS	ICMP Flood	91.2%	High	Time Ban
2026-09-25 12:07:23	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Time Ban (2m)
2026-09-25 12:09:58	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Time Ban (2m)
2026-09-25 12:11:58	10.0.0.19	DDoS	SYN Flood	92.7%	Critical	Time Ban (5m)

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 12:11:58	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Time Ban (5m)
2026-09-25 12:11:58	10.0.0.17	DDoS	SYN Flood	93.4%	Critical	Time Ban (5m)
2026-09-25 12:13:02	10.0.0.4	DDoS	Uncertain	59.6%	Low	Released
2026-09-25 12:13:02	10.0.0.9	DDoS	UDP Flood	92.0%	High	Time Ban (1m)
2026-09-25 12:13:08	10.0.0.10	DDoS	Uncertain	68.4%	Low	Released
2026-09-25 12:13:15	10.0.0.16	DDoS	SYN Flood	92.5%	Critical	Time Ban (2m)
2026-09-25 12:21:51	10.0.0.20	DDoS	UDP Flood	79.4%	High	Time Ban (2m)
2026-09-25 12:22:57	10.0.0.8	DDoS	UDP Flood	92.4%	High	Time Ban (1m)
2026-09-25 12:23:03	10.0.0.1	DDoS	Uncertain	58.9%	Low	Released
2026-09-25 12:23:51	10.0.0.20	DDoS	UDP Flood	79.4%	High	Time Ban (5m)
2026-09-25 12:23:58	10.0.0.8	DDoS	UDP Flood	92.4%	High	Released
2026-09-25 12:24:15	10.0.0.9	DDoS	Uncertain	48.4%	Low	Released
2026-09-25 12:24:26	10.0.0.8	DDoS	UDP Flood	92.4%	High	Quarantined

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.16	SYN Flood	0.6234	93.4%	Critical	Phase 1	2	10s	2026-09-25 12:30:43	Attack Stopped
10.0.0.24	ICMP Flood	0.6455	91.2%	High	Phase 2	1	10s	2026-09-25 12:30:36	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.5988	92.5%	High	Phase 2	2	1m 11s	2026-09-25 12:30:35	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.5983	92.2%	High	Phase 2	2	1m 11s	2026-09-25 12:30:35	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6069	92.7%	High	Phase 2	2	1m 11s	2026-09-25 12:30:33	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.5922	93.4%	High	Phase 2	1	1m 11s	2026-09-25 12:30:33	Ban Expired
10.0.0.23	ICMP Flood	0.6445	90.9%	High	Phase 2	1	10s	2026-09-25 12:30:33	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6454	91.2%	High	Phase 2	1	10s	2026-09-25 12:30:33	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6269	86.7%	High	Phase 2	1	11s	2026-09-25 12:30:04	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	77.5%	High	Phase 2	1	10s	2026-09-25 12:30:03	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6376	92.5%	Critical	Phase 2	1	11s	2026-09-25 12:29:35	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6521	90.1%	Critical	Phase 2	1	11s	2026-09-25 12:29:35	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6506	92.7%	Critical	Phase 2	1	11s	2026-09-25 12:29:33	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6333	91.2%	Critical	Phase 2	1	11s	2026-09-25 12:29:33	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE